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Final Exam
International Campus, MD Program
Nutrition
February 2024 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Hosseinzadeh

Date of Exam: Jun29, 2024

Duration of Exam: 40 Minutes

MCQ Question	40
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



1) Basal Energy Expenditure (BEE) typically representing percent of Total Energy Expenditure (TEE).

- a) 20-30
- b) 31-49
- c) 50-70 ✓
- d) 71-79

2) With increasing age Resting Energy Expenditure (REE) will be:

- a) Increased
- b) Decreased ✓
- c) Calculated by nitrogen loss
- d) Calculated by food intake

3) Which one is different from the others in terms of Thermic Effect of Food (TEF)?

- a) Diet-induced thermogenesis ✓
- b) Non exercise activity thermogenesis
- c) Specific dynamic action
- d) Specific effect of food

4) Which one is correct regarding to BMI?

- a) In healthy weight person BMI is (18.5-24.9) ✓
- b) In healthy weight person BMI is (25-29.9)
- c) In obesity BMI is (40-50)
- d) In extreme obesity BMI is over 30

5) In elderly the need of protein and daily energy will:

- a) Increase for both
- b) Decrease for both
- c) Increase and decrease respectively ✓
- d) Decrease and increase respectively

6) Which of the following produces 4 Kilocalorie per gram?

- a) Saturated fatty acids
- b) Alcohol
- c) Whole grains ✓
- d) Omega 3 fatty acids

7) In elderly who are complaining of muscle pain which vitamin deficiency is more prominent?

- a) Vit A
- b) Vit C
- c) Vit D ✓
- d) Vit B6 ✓

8) Which micronutrient deficiency is more related to delay wound healing and pressure sore development in elderly?

- a) Iron
- b) Iodine
- c) Copper
- d) Zinc ✓

9) Which one contains more levels of mono unsaturated fatty acid (MUFA)?

- a) Olive oil ✓
- b) Sunflower oil
- c) Canola oil
- d) Coconut oil

10) Which one could be a better ratio of Omega-6:Omega-3 for our health?

- a) 32:1
- b) 16:1
- c) 8:1
- d) 4:1 ✓

11) In terms of the definition of malnutrition which one is the best answer?

- a) Malnutrition results from low intake of nutrients compared to the body's needs
- b) Malnutrition results from imbalance between the intake of nutrients and the body's needs ✓
- c) Malnutrition results from high intake of energy compared to the body's needs
- d) Malnutrition results from low intake of protein and energy compared to the body's needs ✓

12) Which food in cholesterol free?

- a) Chicken
- b) Egg
- c) Coconut ✓
- d) Lean meat

13) In Indirect Calorimetry (IC) which items will be quantified?

- a) individual's nitrogen consumption and carbon dioxide production
- b) individual's oxygen consumption and carbon dioxide production ✓
- c) individual's oxygen consumption and nitrogen production ✓
- d) individual's nitrogen consumption and hydrogen production

14) Which factor is hindering calcium absorption?

- a) Acidic environment
- b) Calcitriol
- c) Low intake
- d) Vitamin D deficiency ✓



15) In obesity how many Kilocalories of energy per day should be restricted to lose one pound of body fat in a week?

- a) 500 ✓
- b) 1000
- c) 1500
- d) 2000

16) Which clinical sign could be more related to kwashiorkor compared to Marasmus?

- a) Frequent infection
- b) Electrolyte imbalance
- c) Hair changes ✓
- d) Frequent dehydration

17) Which item in Healthy Eating Plate has been considered as the most area of the plate?

- a) Whole grains
- b) Healthy proteins
- c) Fruits
- d) Vegetables ✓

18) Which one is digested by colonic bacteria?

- a) Maltose
- b) Lactose
- c) Fructose
- d) Raffinose ✓

19) Which part of GI has no enzyme to digest carbohydrate?

- a) Mouth
- b) Stomach ✓
- c) Small intestine
- d) Pancreas

20) Which one is facilitated transporter for Fructose?

- a) GLUT 1
- b) GLUT 2
- c) GLUT 4
- d) GLUT 5 ✓

21) Which part of amino acid does dedicate the amino acid function?

- a) Alpha carbon
- b) R group ✓
- c) Hydrogen
- d) Amino

Which one is the most common limiting amino acid?

- a) Lysine ✓
- b) Tyrosine
- c) Phenylalanine
- d) BCAA

23) Select the symptom(s) that characterize thiamin toxicity.

- a) Sensory neuropathy
- b) You can't consume enough thiamin for it to be toxic. ✓
- c) Nausea, jaundice and elevated liver enzymes.
- d) Numbness or tingling in the hands, arms and face.

24) Which of the following conditions is described as vitamin D deficiency in adults that is characterized by a softening of the bones?

- a) Osteomalacia ✓
- b) Osteoarthritis
- c) Osteoporosis
- d) Osteosarcoma

25) We know that vitamin C is a water-soluble vitamin. Therefore, which of the following is a TRUE statement?

- a) Vitamin C is easily stored in your body for long periods of time.
- b) It is easy to consume too much vitamin C and suffer from toxicity.
- c) Vitamin C does not need to be replenished regularly.
- d) Vitamin C is easily flushed out of your body. ✓

26) What is the term used to describe a type of anemia in which the red blood cells are large and immature, reducing the blood's ability to transport oxygen?

- a) Microblastic anemia
- b) Hemolytic anemia
- c) Megaloblastic anemia ✓
- d) Pernicious anemia

27) Which of following populations are not at risk of vitamin D deficiency?

- a) Breast-fed infants
- b) Individuals with high body mass index
- c) Patients with chronic kidney disease
- d) Patients with osteopenia ✓

28) For a pregnant women with pre-pregnancy BMI of 28 kg/m^2 the recommended total weight gain is?

- a) 12-18 kg
- b) 11-16 kg
- c) 7-11 kg ✓
- d) 4.5-9 kg



29) The intake of which food groups are recommended to be increased in pregnancy compared pre-pregnancy?

- a) Fruits and vegetables
- b) Dairy and meats ✓
- c) Cereals
- d) Fats and sweets

30) Which nutrient in colostrum is higher than mature breast milk?

- a) Carbohydrate
- b) Fat
- c) Protein ✓
- d) Lactose

31) Which foods should be avoided for infants during the entire first year of life?

- a) Apple juice
- b) Rice
- c) Egg whites ✓
- d) Potato

32) Which of the supplements should be given during the first 2 years of life?

- a) Vitamin A, Vitamin D, Vitamin C
- b) Vitamin A, Vitamin D, Zinc
- c) Vitamin A, Vitamin D, Iron ✓
- d) Vitamin D, Vitamin C, Zinc

33) Which of the following options indicates severe malnutrition in the context of injury or acute illness?

- a) Energy intake $\leq 50\%$ for ≥ 5 days ✓
- b) Energy intake $\leq 70\%$ for < 5 days
- c) Energy intake $\leq 60\%$ for < 7 days
- d) Energy intake $\leq 40\%$ for ≥ 5 days

34) Deficiency of which of the following micronutrients can cause petechiae?

- a) Vit C ✓
- b) Vit A
- c) Iron
- d) Zinc

35) Flaky nails can be caused by which of the following conditions?

- a) EFAD ✓
- b) Iron deficiency
- c) Cu deficiency
- d) Hypokalemia

36) The benefit of MCT Oil that add to lipid emulsion is:

- a) It is an essential fatty acid
- b) It prepares high calorie
- c) It is a Hydrolyzed oil.
- d) It does not require bile salts or pancreatic lipase for digestion ✓

37) Fructooligosaccharides (FOS), which have been added to enteral formulas, have these effect, except:

- a) Changes in colonic pH
- b) Changes in fecal microbiota
- c) Changes in bowel movements ✓
- d) Changes in short-chain fatty acid concentrations ✓

38) Which of the following ions does not change in Refeeding syndrome?

- a) Na ✓
- b) Mg
- c) P
- d) K

39) Which of the following is not correct about lipid emulsion in parenteral nutrition?

- a) Administration should not exceed 2 g of lipid emulsion per kilogram of body weight per day
- b) The lipid infusion should be discontinued When triglyceride level exceeds 400 mg/dl
- c) The PN most often is infused during 8 hours per day with the lipid as part of the PN solution ✓
- d) 10% lipid emulsion contains 1.1 kcal/ml

40) Which of the following is not related to signs and symptoms to gastric feeding?

- a. Distension
- b. Diarrhea
- c. Aspiration ✓
- d. Constipation ✓

2. The answers should be marked with a dot and dark pencil in the appropriate box.

Please don't write anything in this box.

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